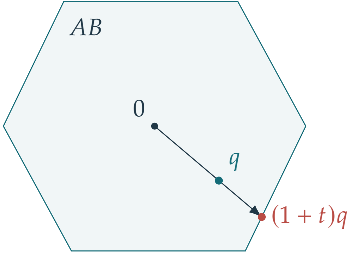
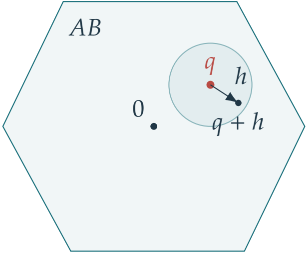
Radial extension from an interior point



$$Au = (1+t)q,\ u \in B$$
$$z = ru/(1+t) \text{ gives}$$
$$Az = Ax_0 \text{ and } \|z\|_1 < r.$$

A better feasible vector gives an interior point



$$Az = Ax_0,\ \|z\|_1 < r$$
$$q + h = A(z/r + A^+h) \in AB$$
$$\text{for small } h.$$

$$r = \|x_0\|_1 > 0, \qquad q = Ax_0/r, \qquad A \text{ onto}$$